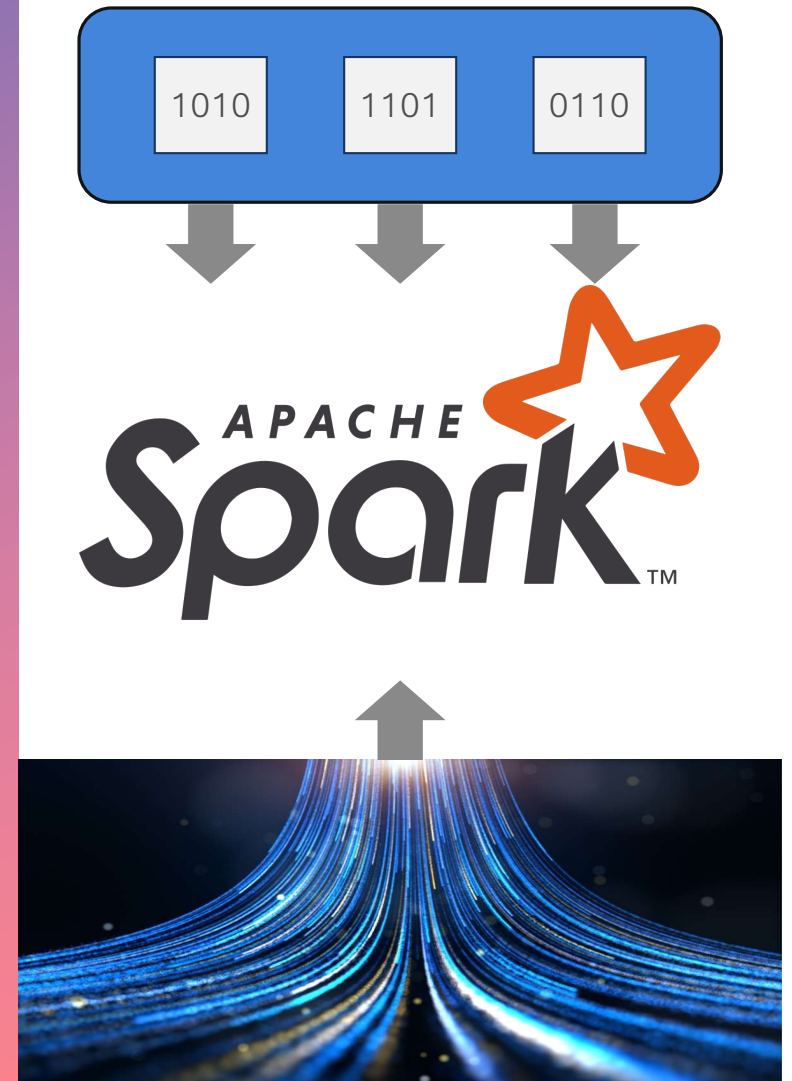


Apache Spark

Introduction

www.3aayaam.in

1



Agenda

1. What is Spark
2. Spark in Data Engineering
3. Batch & Stream Processing
4. Spark Ecosystem
5. Spark Basic Architecture
6. PySpark
7. Install, Setup & Configure Spark
8. Setup PySpark
9. Spark CLI
10. Spark UI

What is Spark

- Open-source **Distributed Data Processing** Framework.
- Spark and Hadoop are two of the top Apache projects.
- Big data processing and data analytics for large scale data.
- Python, Java, Scala, R, SQL.
- Batch and Stream processing.
- Parallel Processing.
- Most important technology in the Big Data Ecosystem and Data Engineering.

Spark in Big Data Ecosystem

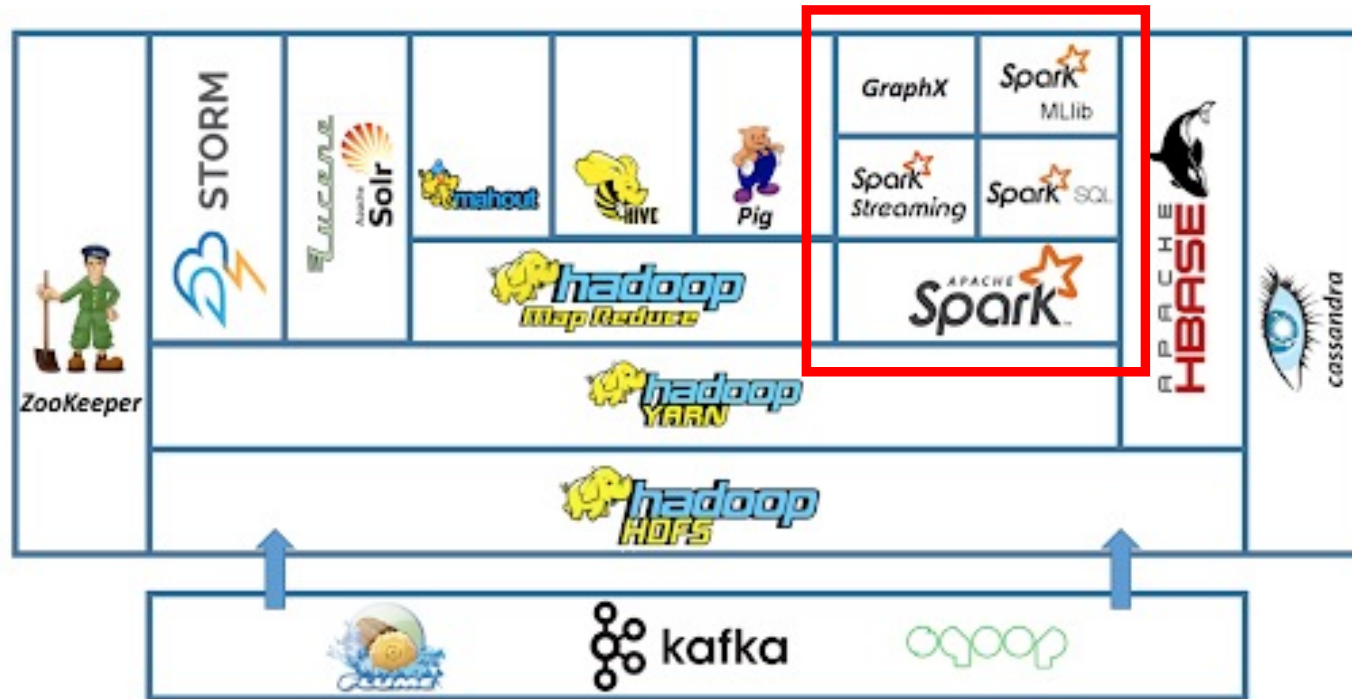
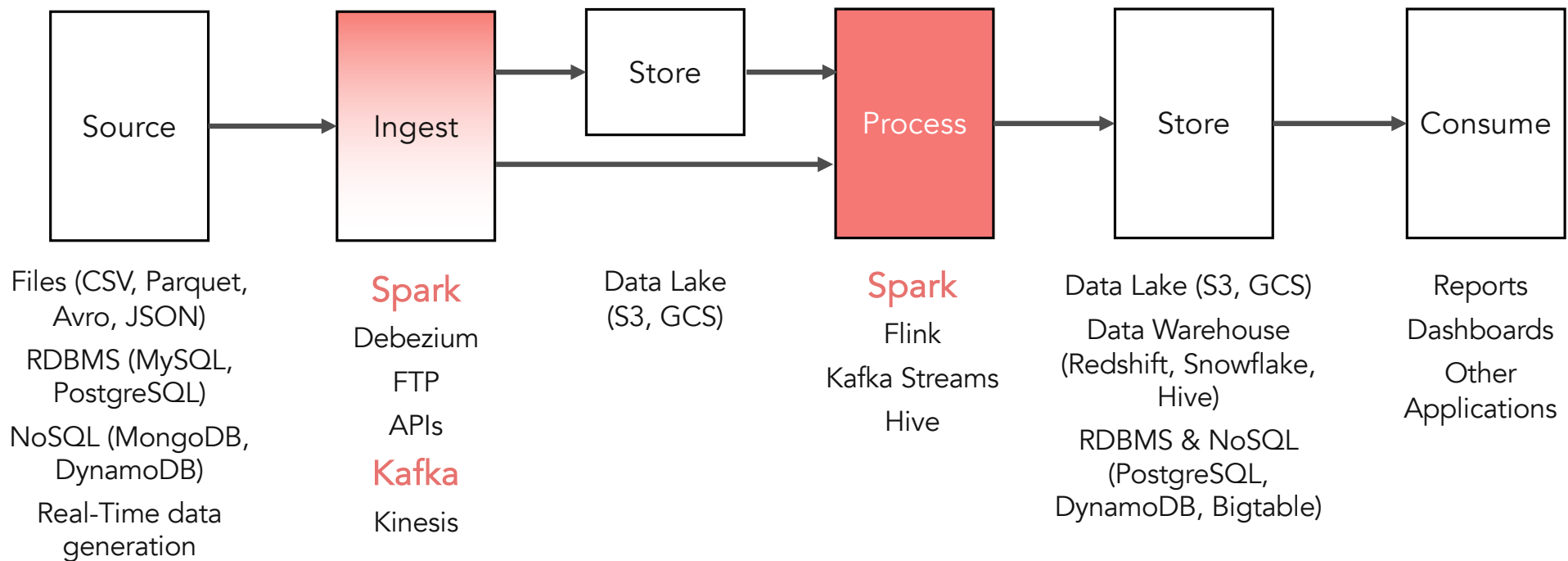


Image Courtesy: Google Search

Spark in Data Engineering

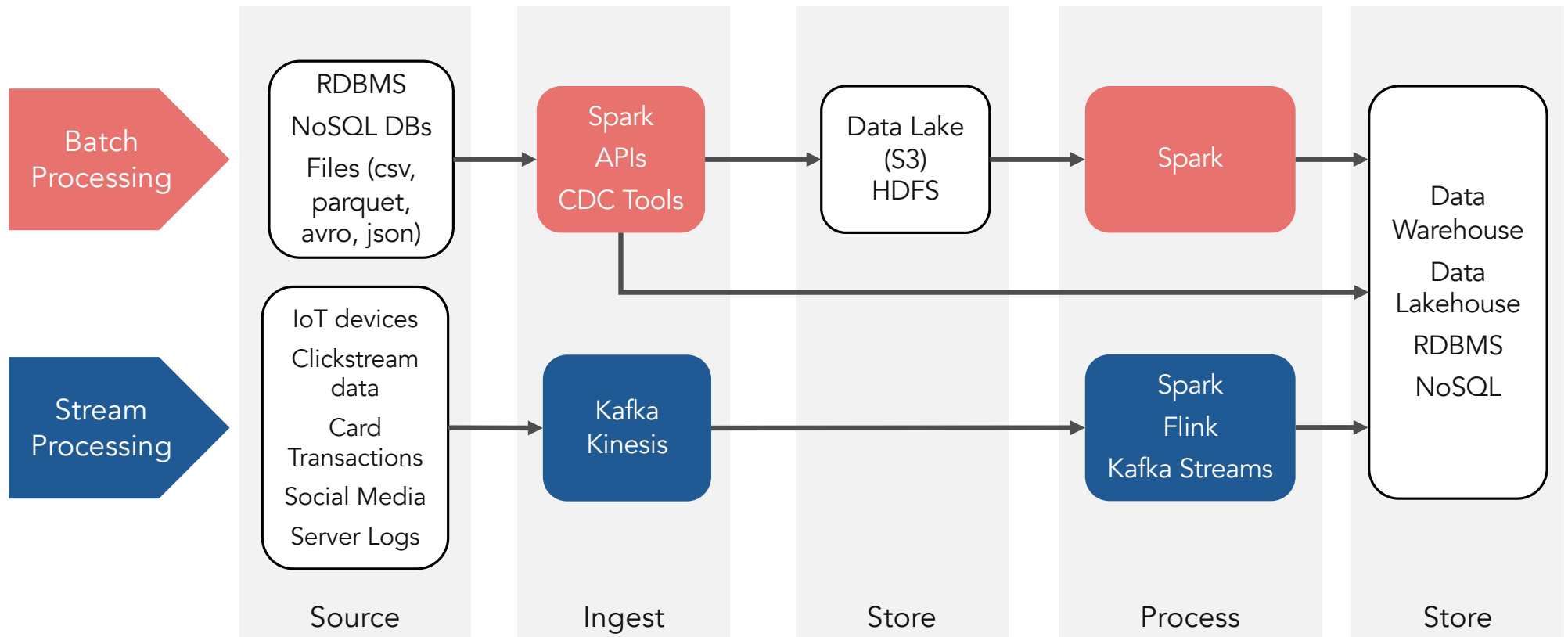


Batch and Stream Processing

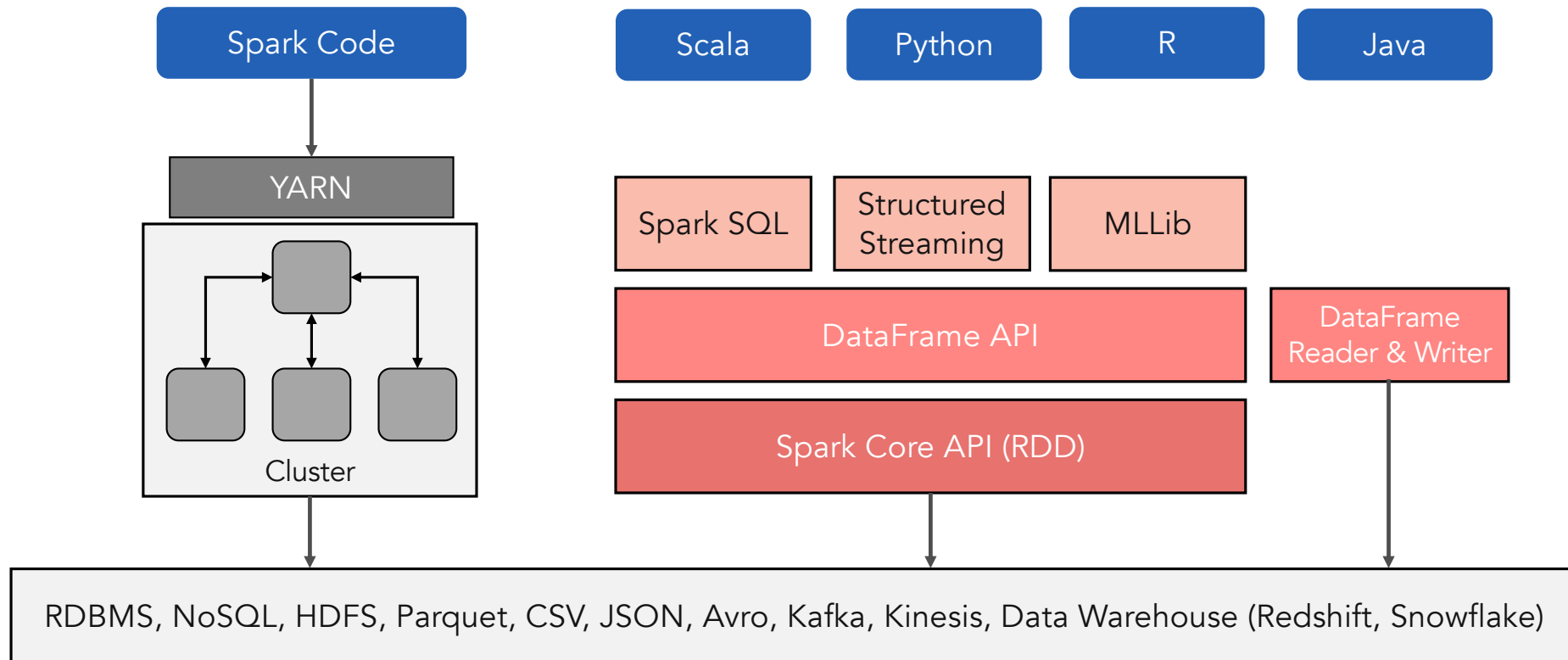
- What is Batch ?? Collection of large volume of finite data that is processed together.
 - Collection – Accumulation of certain size of data or no. of records over a specified period of time. E.g. a few hours, day/days, week/weeks, month/months etc.
 - Finite and Bounded data – We know the size of data or no. of records or data with proper beginning and end etc.
 - Longer data relevance.
- Streaming Data??
 - Large volume of data that is continuously generated.
 - Infinite and Unbounded.
 - Real-time or near real-time processing.
 - Shorter data relevance.

Processing of batch data is Batch Processing.
Processing of streaming data is Stream Processing.

Batch and Stream Processing

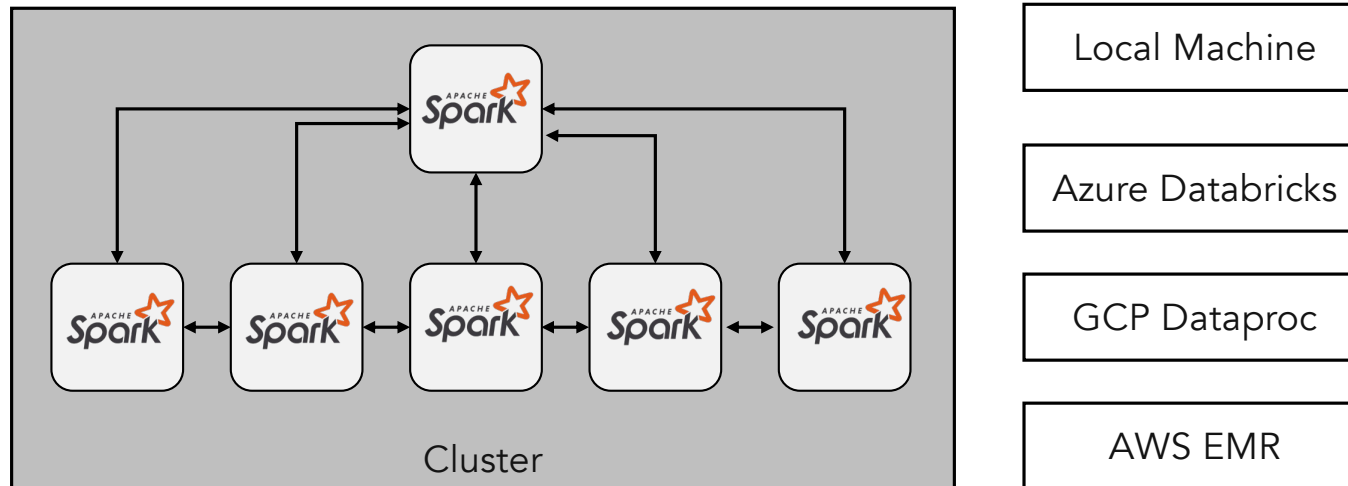


Spark Ecosystem



Hosting Platforms for Spark

- What is a hosting platform?
- Used to host a tool, framework, application, software etc.
- AWS EMR, GCP Dataproc, Databricks (AWS/GCP/Azure).

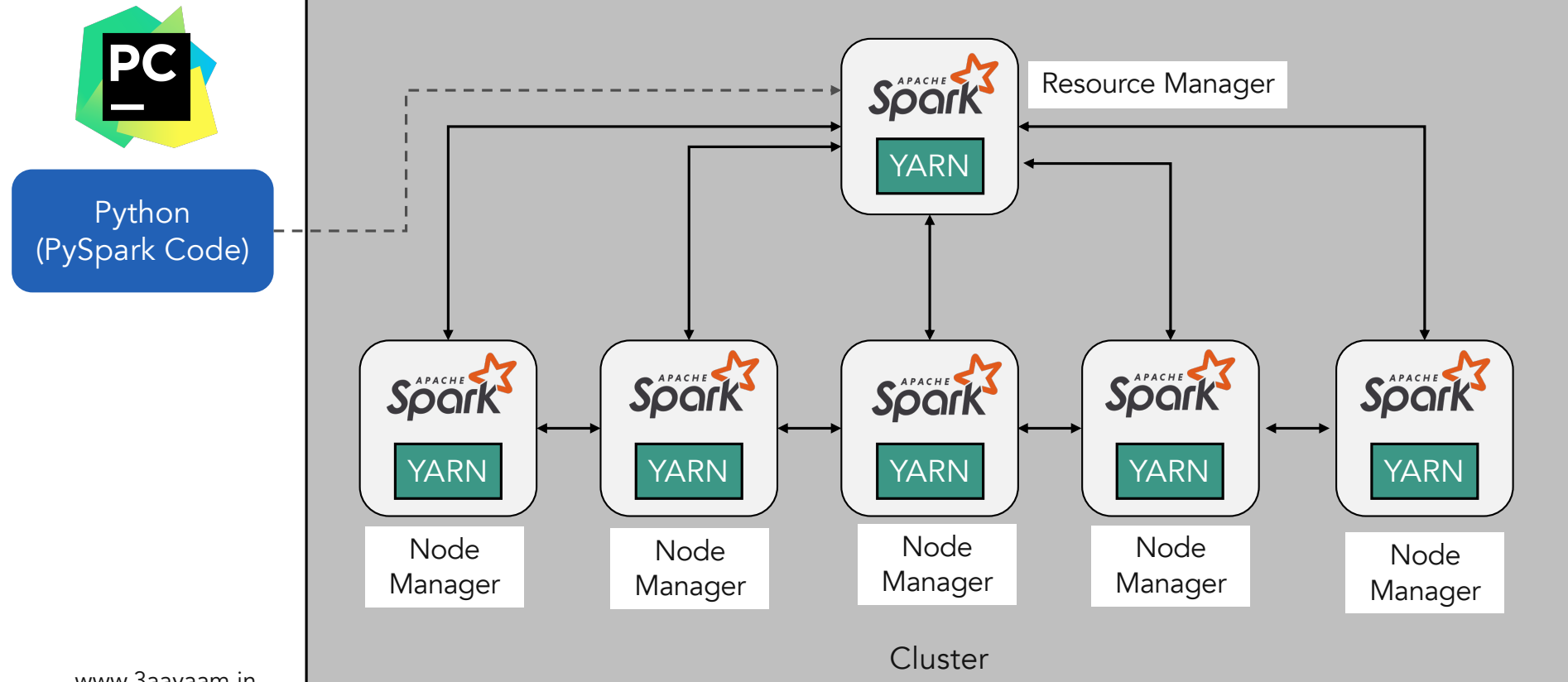


Spark Development

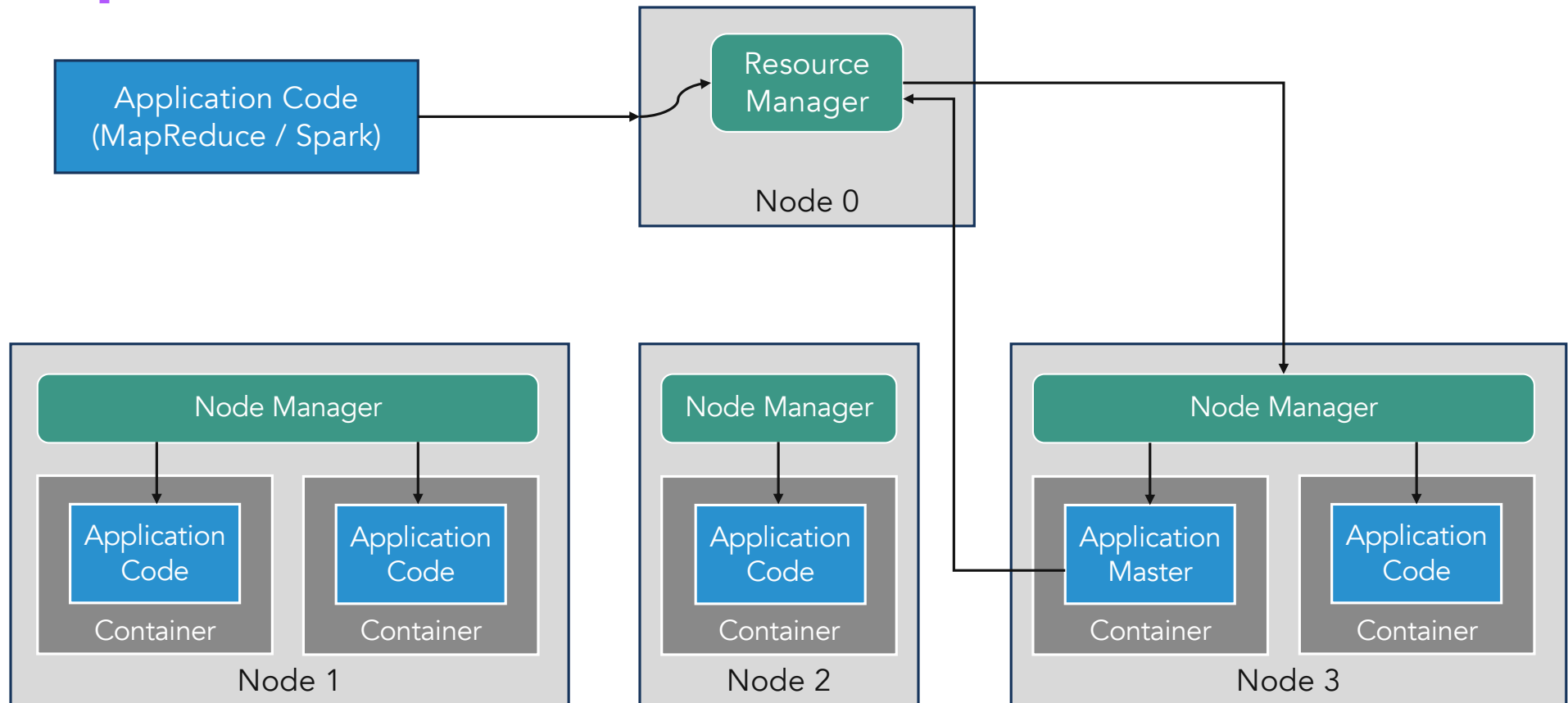
- Python
- Scala
- Java
- R



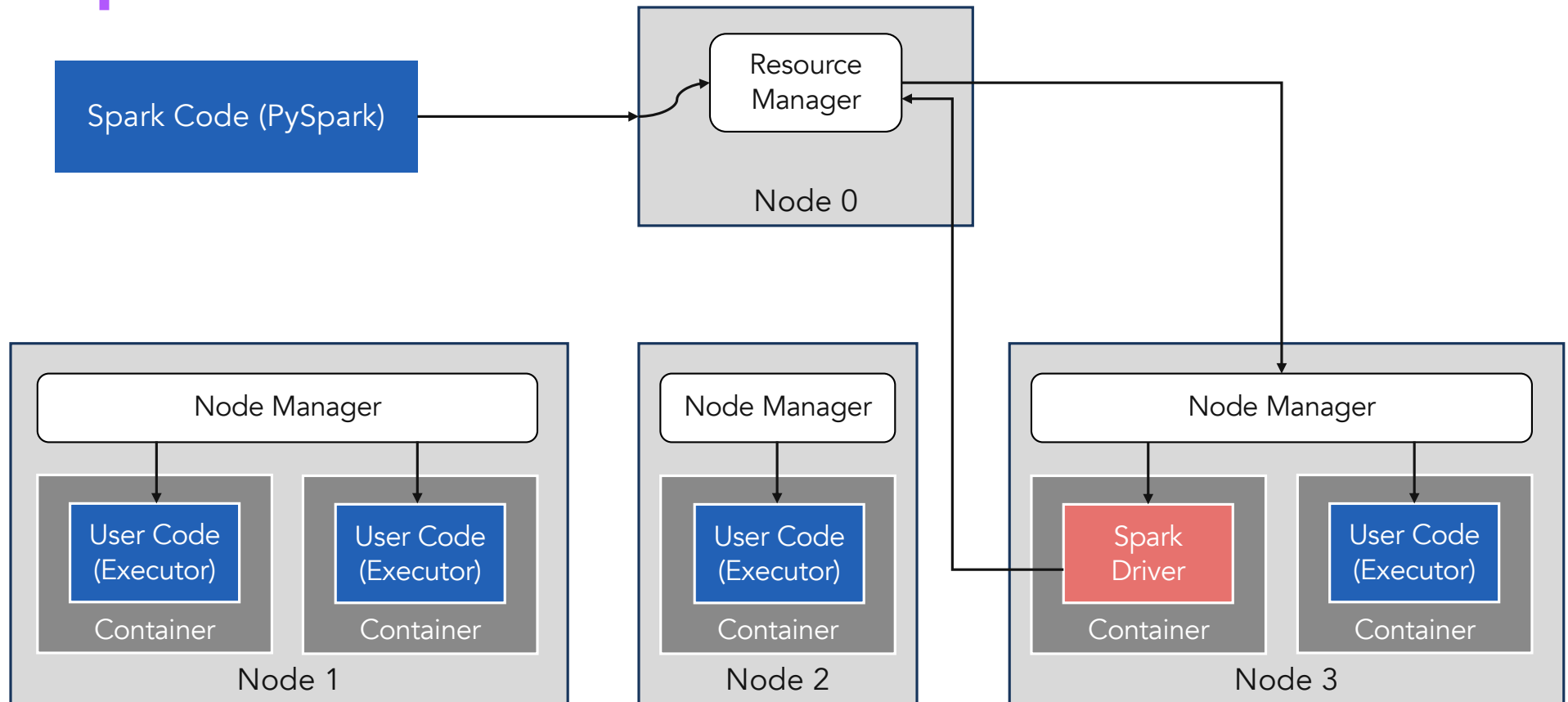
IDE + Spark + YARN + Cluster



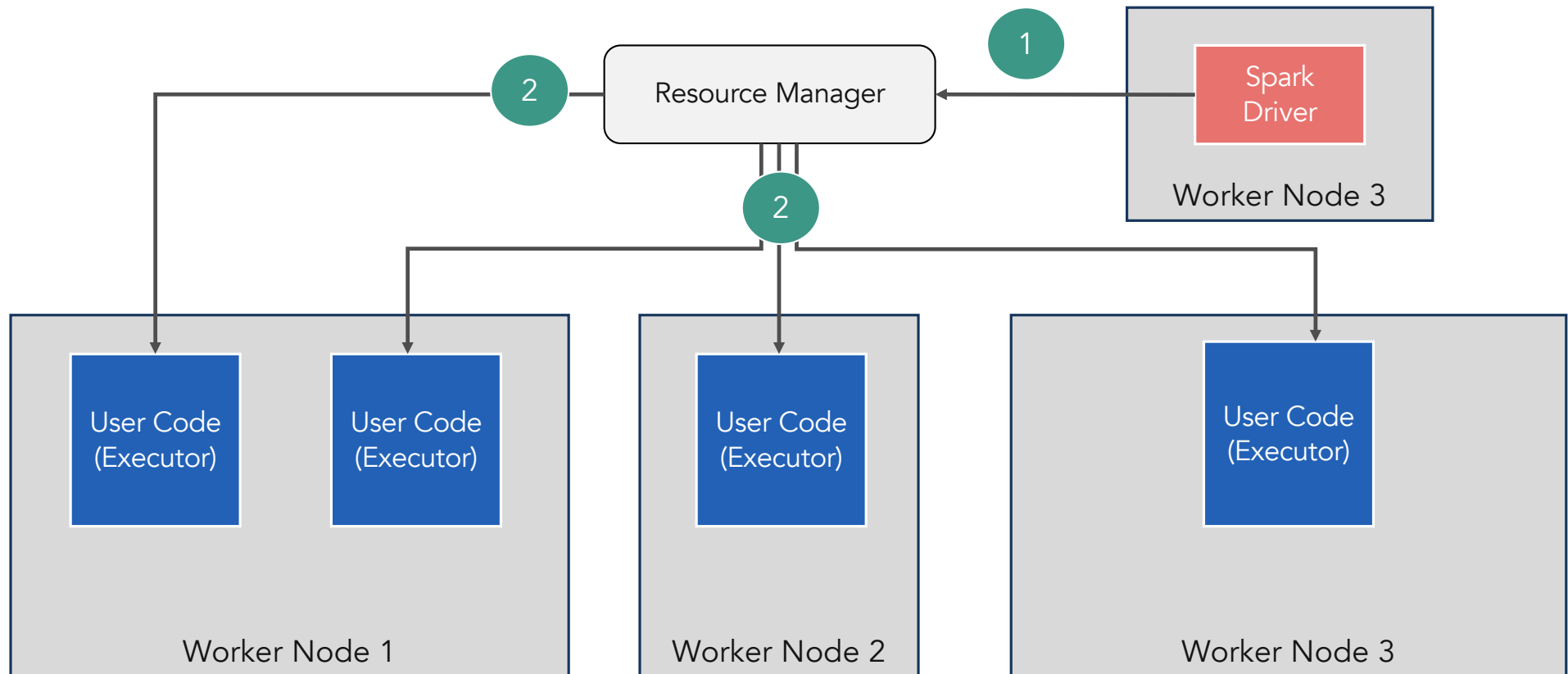
Spark Execution Architecture



Spark Execution Architecture



Spark Architecture



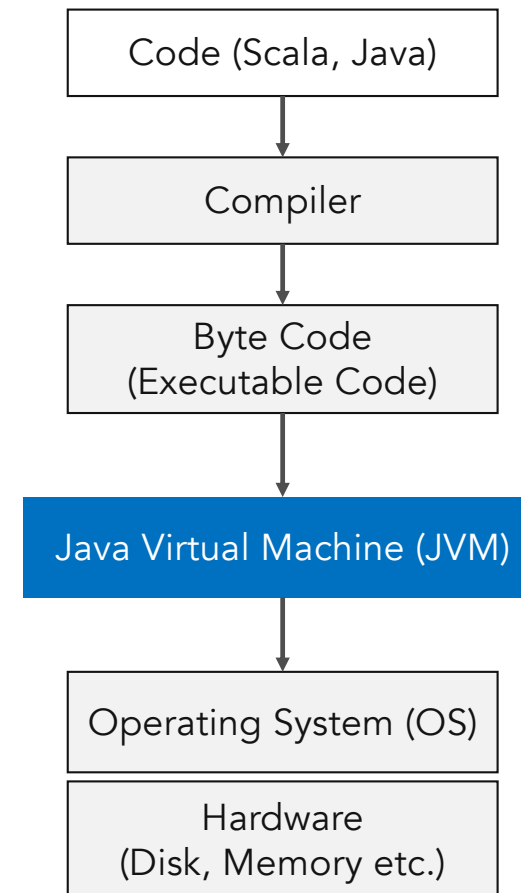
Spark Architecture

- Driver Process
 - Coordinates with Resource Manager for compute resources.
 - Analyzes the code and creates execution plan.
 - Schedules jobs and tasks.
 - Tracks the status of the jobs. Failed tasks are replaced with new tasks by Driver.
- Executor Process
 - Runs the actual transformation code based on the access plan provided by Driver.

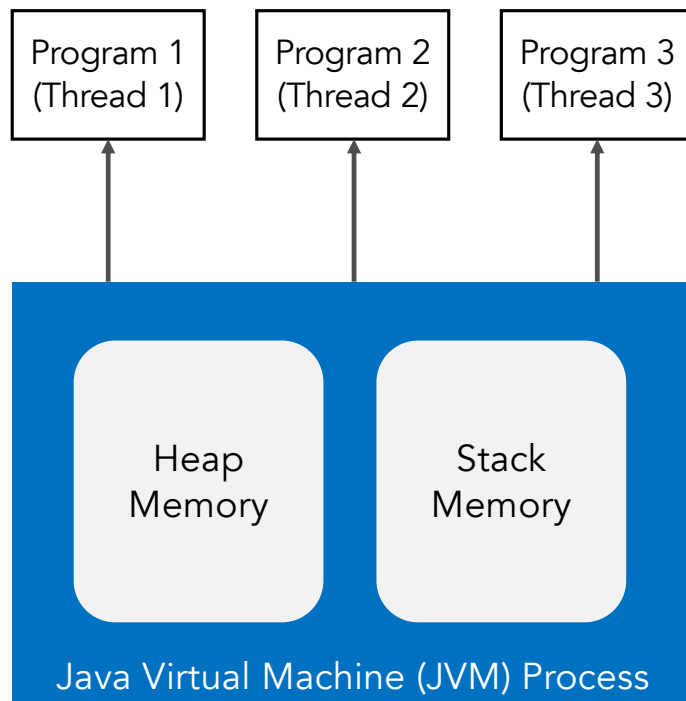
Set Up

Spark Codebase

- Spark is written in Scala.
- Scala is JVM based programming language.
- Java, Kotlin, Clojure.
- What is meant by JVM based ??
 - Take compiled code as input and execute them using the server resources.



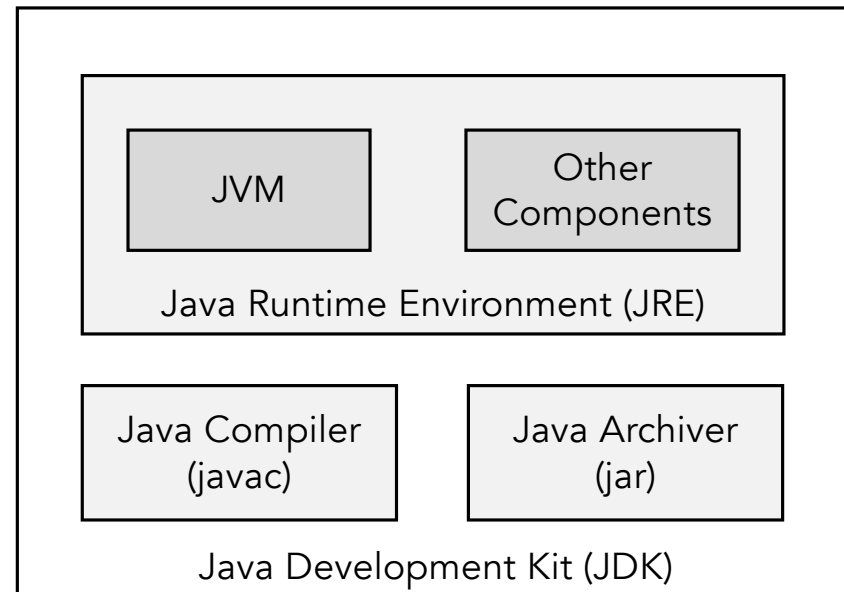
A bit more on JVM



- A single server can have one or more JVMs running.
- Executables/programs run as threads on JVM. All of them share the heap and stack memory.
- Each executable will have its own stack.
- Heap memory – Used to store variables, objects etc.
- Stack memory – Used by methods to store local variables and can refer objects and variables in Heap memory.

Installation Steps

- Requirements:
 - JVM.
 - Scala.
 - Spark.
- Steps
 - Download & Install JDK
 - Download & Install Scala
 - Download & Install Spark
 - Set up environment variables
- Let's Install Spark

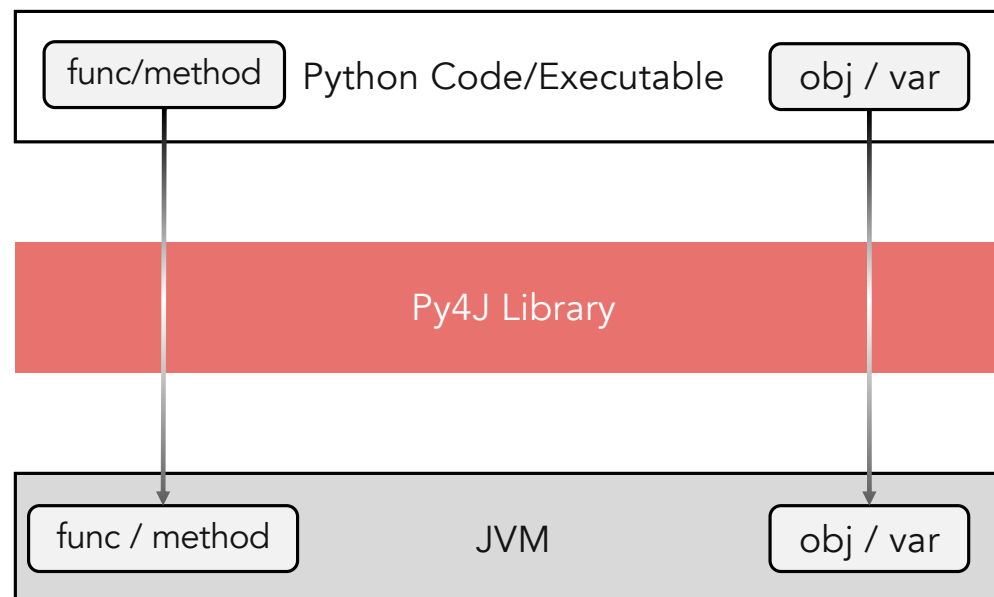


Configuration

- `spark.eventLog.enabled`
- `spark.eventLog.dir`
- `spark.history.fs.logDirectory`
- Start history server using script present in `SPARK_HOME/sbin/start-history-server.sh`
- History server is available in `localhost:18080`
- PySpark set up
 - Open PyCharm. Go to "Run", then "Edit Configurations".
 - Set up "Environment Variables" with below:
 - `SPARK_HOME = Spark Folder`
 - `PYTHON_PATH = $SPARK_HOME/python:$SPARK_HOME/python/lib/py4j-0.10.9.7-src.zip`

PySpark

- Python Library for Spark.
- pip3 install pyspark.
- Py4J gets installed with PySpark.
- CLI and Spark UI



Summary

- Spark Introduction
- Fitment of Spark in Data Engineering
- Batch & Stream Processing
- Spark Ecosystem
- Hosting Platforms
- Spark Development (IDE, Cluster, YARN)
- Spark Execution Architecture
- Spark Architecture
- PySpark, Py4J and Spark
- Installation & Configuration on Laptop



Thank You
www.3aayaam.in